Test 10.3 **Marks: 50**

Graphs (straight lines) **Time: 60 min.**

QUESTION 1

1.1 The x- and y-values in the table below represent the coordinates of three points.

x	1	2	3
y	5	7	10

Explain (without plotting the points) why the points will not form a straight line. (2)

1.2 The table below represents points that lie on a straight line.

x	-2	2	a	7
y	8	-8	-16	b

1.2.1 Write down the gradient of the line. (2)

1.2.2 Find the values of a and b in the table. (4)

[8]

QUESTION 2

2.1 Will the point $(-4; -4)$ lie on the line with the equation $y = -\frac{1}{2}x - 2$?

Explain your answer. (2)

2.2 Find the value of p if the point $(p; -6)$ lies on the line with the equation $y = -\frac{1}{2}x - 2$. (2)

2.3 Will the line with the equation $4y + 8x = -8$ be perpendicular to the line with the equation $y = -\frac{1}{2}x - 2$? Give a reason for your answer. (3)

[7]

QUESTION 3

Find the equations of the following straight line graphs:

3.1 the line which intersects the x-axis at 3 and the y-axis at 6 (3)

3.2 the line which is parallel to the line in **3.1** above and passes through the point $(-4; 10)$ (3)

3.3 the line which is perpendicular to the y-axis and passes through the point $(-4; 10)$ (2)

[8]

QUESTION 4

The equation of line f is $y = \frac{1}{3}x - 2$, the equation of line g is $y = -2x$ and the equation of line h is $y = 4$.

4.1 Draw sketch graphs of f, g and h on the same set of axes. (Clearly indicate all intercepts with the x- and y-axis on your sketch.) (8)

4.2 Find the points of intersection of the following lines:

4.2.1 f and g (4)

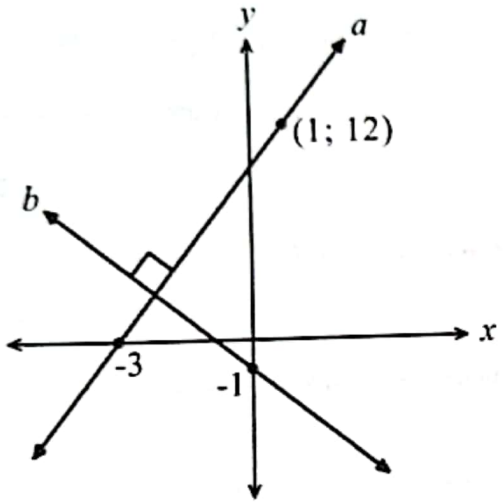
4.2.2 g and h (2)

[14]

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QUESTION 5

5.1

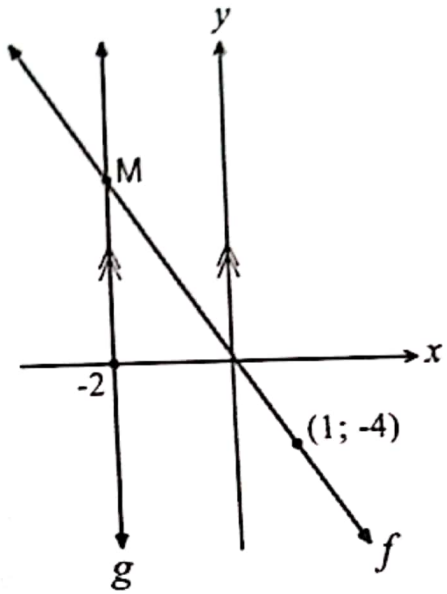


Use the information in the diagram to find the following:

5.1.1 the equation of line a (4)

5.1.2 the equation of line b (2)

5.2



Use the information in the diagram above to find the following:

5.2.1 the equation of line f (3)

5.2.2 the equation of line g (2)

5.2.3 the coordinates of point M (2)

[13]

TOTAL: 50

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